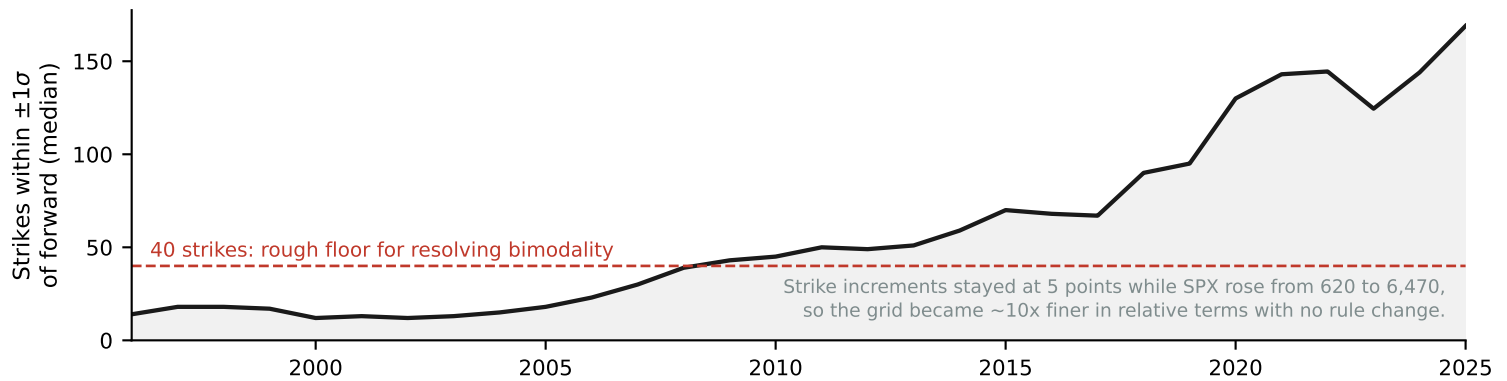


SPX option chain: strike density diagnostics

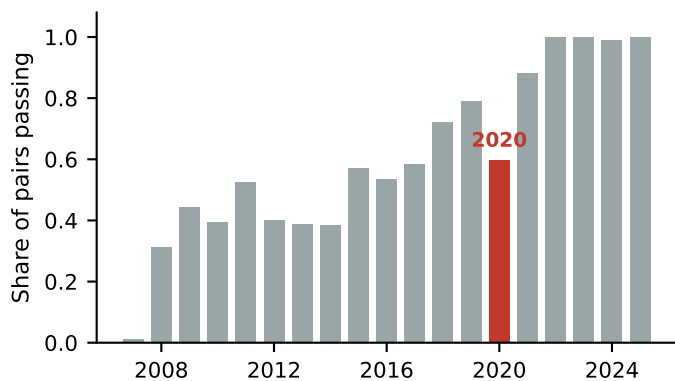
OptionMetrics IvyDB US, secid 108105, AM-settled monthlies only.

1996-01-04 to 2025-08-29, 18,046,547 rows. Date-expiry pairs with 14 to 90 days to maturity.

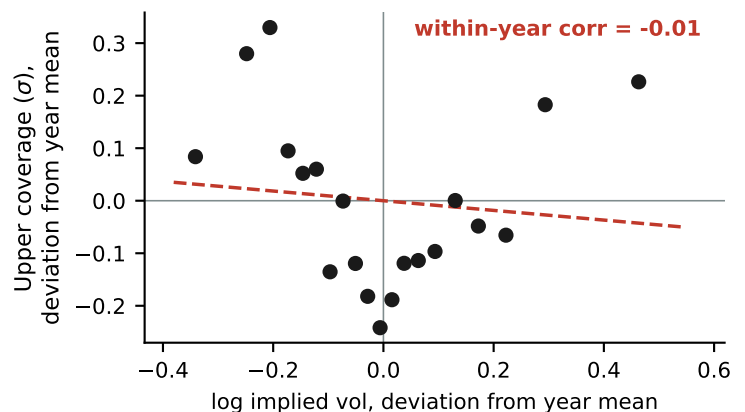
A. Strike density has risen ~12x



B. Pass rate, 2007 onward



C. No volatility dependence



Year	Strikes	Within $\pm 1\sigma$	ATM gap %	Upper σ	Lower σ	Median IV	Pass
1996	29	14	0.76	+1.2	-5.4	0.171	0.00
2000	29	12	1.64	+2.3	-3.9	0.230	0.00
2005	46	18	0.42	+2.9	-11.5	0.147	0.00
2008	101	39	0.40	+3.4	-6.8	0.250	0.31
2010	150	45	0.45	+3.4	-15.1	0.288	0.39
2015	229	70	0.24	+3.3	-17.9	0.242	0.57
2018	268	90	0.18	+3.4	-14.7	0.224	0.72
2020	346	130	0.16	+3.2	-23.1	0.343	0.60
2022	398	144	0.12	+6.1	-30.3	0.261	1.00
2025	428	169	0.08	+7.8	-39.2	0.226	1.00

Coverage is measured in units of $\sigma\sqrt{T}$, not raw moneyness: a fixed moneyness band spans very different probability mass across maturities and volatility regimes. In 1996 the highest listed strike sat only $\sim 1.2\sigma$ above the forward, so the entire right tail of the risk-neutral density would be extrapolation rather than observed prices, and any entropy measure would largely reflect the tail-fitting assumption rather than market prices.

Panel C tests whether sigma-unit coverage is endogenous to volatility. It is not: within year, upper coverage correlates -0.01 with log implied vol, and the pass rate itself +0.03. The listed strike range does widen with volatility (corr +0.28), offsetting the mechanical effect of implied vol entering the denominator.

Proposed approach: retain every observation and decompose the entropy integral into an observed component and a tail-extrapolated component, reporting the extrapolated probability share as a diagnostic and testing robustness across its terciles. Decomposition is preferred not because screening is biased, but because it converts a tail-extrapolation assumption into a measured and reportable quantity.